A variational spectral lower bound from concentration of separated sets

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3 November 2026

Abstract

We derive a quantitative variational spectral lower bound from the multiset concentration hypothesis in AIM 1013, under nonnegative Bakry–Émery curvature. The proof uses a half-volume isoperimetric region and separated distance layers, and applies without assuming discrete spectrum.

Repository target. This manuscript concerns AIM problem 1013 at revision `3fa50884d129`. Its scope is stated below.

Result

The answer to Problem 1013 is **yes**. In fact its concentration hypothesis implies a lower bound for the first variational gap, and hence for every higher variational level:

$$\boxed{\lambda^{(k)} \geq \lambda^{(1)} \geq \frac{a}{64k^2(k-1)^2\log 2}, \qquad k \geq 2.} \tag{1}$$

Here a is exactly the parameter in the question. The constant is not claimed to be optimal. The proof applies to complete, possibly noncompact, smooth connected weighted Riemannian manifolds and does not require discrete spectrum.

The geometric input is the classical isoperimetric comparison theory under $\mathrm{Ric} + \mathrm{Hess}\, V \geq 0$. The additional step is to divide one side of a half-volume isoperimetric region into alternating pieces and gaps of prescribed measures. The other half of the manifold serves as one of the k pieces.

1. Isoperimetric facts and their precise scope

Write

$$P_\mu(E) = \int_{\partial^* E} e^{-V}\, d\mathcal{H}^{n-1}, \qquad \mathcal{I}(v) = \inf_{\mu(E)=v} P_\mu(E), \qquad I = \mathcal{I}(1/2),$$

where $\partial^* E$ is the reduced boundary and $n = \dim M$. Under the assumptions of the question, the following facts apply.

1. Isoperimetric minimizers of every measure $0 < v < 1$ exist. A minimizer can be represented by an open set. Its boundary is smooth outside a singular set of Hausdorff dimension at most $n - 8$, and its regular part has constant weighted mean curvature. Its weighted perimeter is the weighted area of that regular part.
2. The profile $\mathcal{I}$ is symmetric about $1/2$ and concave on $(0, 1)$. Consequently its Cheeger constant is
$$h_\mu := \inf_{0<\mu(E)<1} \frac{P_\mu(E)}{\min(\mu(E), 1 - \mu(E))} = 2\mathcal{I}(1/2) = 2I. \tag{2}$$
3. Normal rays from the regular boundary of an isoperimetric region cover its interior up to a set of measure zero. Weighted normal Jacobians obey the Heintze–Karcher comparison estimate under the Bakry–Émery curvature bound.

These facts hold for the complete finite-measure setting, not just for compact manifolds. In particular, finite total measure prevents loss of a prescribed positive volume to infinity in the existence argument: local BV compactness, followed by the uniform bound $\mu(M \setminus K) \to 0$ along a compact exhaustion, gives global convergence in $L^1(\mu)$. Lower semicontinuity of weighted perimeter then gives a minimizer. Boundary regularity and the normal-ray assertion are the standard geometric measure theory and comparison inputs cited below.

For (2), concavity and nonnegativity imply

$$\mathcal{I}(v) \geq 2v\mathcal{I}(1/2) \quad (0 < v \leq 1/2).$$

Symmetry gives the corresponding inequality for $v \geq 1/2$, while a half-volume minimizer gives equality in the infimum defining h_μ.

The inputs can be found together in Milman (2010), Sections 2–3: existence and regularity in Section 2.2, the normal-ray treatment of singular boundaries in Section 2.4, and Theorem 3.2 and Corollary 3.3 for the profile and Cheeger constant. Milman (2015), Theorem 2.3, Corollary 2.5, and Proposition 2.9 give a detailed treatment of the geometric inputs, including the fact that nearest boundary points are regular. The required comparison is a specialization of its Theorem 2.1.

These statements do not require an orientation of the ambient manifold. Weighted perimeter uses the Riemannian density; an isoperimetric region supplies an outward normal on its regular boundary, even if the ambient manifold is nonorientable. The compactness, regularity, first-variation, and normal-Jacobian arguments are local and apply in coordinate charts without an ambient orientation.

The one-dimensional cases are also included. A complete connected one-dimensional manifold without boundary is a line or a circle. On the line the density is log-concave, and half-lines are isoperimetric minimizers; on a circle, $V'' \geq 0$ forces V to be constant and arcs are minimizers. Their distance-density comparisons below follow directly from log-concavity, or from the same normal-ray calculation with no tangential second fundamental form. The Cheeger-profile identity (2) holds in these cases as well.

2. A half-volume region with bounded inward distance density

Choose an open isoperimetric region E with

$$\mu(E) = \frac{1}{2}, \qquad P_\mu(E) = I.$$

Its complement, represented by its interior, is another half-volume minimizer with the opposite weighted mean curvature. Thus we may choose the side so that the weighted mean curvature

with respect to its outward normal is nonnegative:

$$H_\mu(E) = \operatorname{div}_{\partial E} \nu - \langle \nabla V, \nu \rangle = H \geq 0.$$

Here the curvature is the trace of the second fundamental form, without division by $n-1$, minus the outward normal derivative of V. Define the globally 1-Lipschitz function

$$f(x) = d(x, M \setminus E). \tag{3}$$

It vanishes on the complement and is positive in E. We claim that, for all $0 \leq s < t < \infty$,

$$\mu\{s < f < t\} \leq I(t-s). \tag{4}$$

For clarity, (4) is a bound on every distance layer, stronger than a bound only on the total layer adjacent to the boundary.

To prove it, let S be the regular part of ∂E, and follow the inward unit normal from $x \in S$ until its first cut time $\tau(x)$. Let $J_x(t)$ be the normal Jacobian including the density ratio, normalized by $J_x(0) = 1$. Before the cut time, the Riccati equation gives

$$\frac{d^2}{dt^2} \log J_x(t) = -|B_t|^2 - \big(\operatorname{Ric} + \operatorname{Hess} V\big)(\dot\gamma, \dot\gamma) \leq 0.$$

The initial logarithmic derivative is $-H$. Hence

$$J_x(t) \leq e^{-Ht} \leq 1, \qquad 0 \leq t < \tau(x). \tag{5}$$

The normal-ray representation, including the standard treatment of the singular boundary and cut locus, now yields

$$\begin{aligned} \mu\{s < f < t\} &\leq \int_S \int_s^t \mathbf{1}_{\{r < \tau(x)\}} J_x(r)\, dr\, e^{-V(x)} d\mathcal{H}^{n-1}(x) \\ &\leq (t-s) P_\mu(E) = I(t-s). \end{aligned}$$

This proves (4). Equivalently, the pushforward of $\mu|_E$ under f is absolutely continuous on $(0, \infty)$, with density at most I, and total mass $1/2$. The quantity I is finite and positive: a set of zero weighted perimeter on a connected manifold with positive smooth density is constant almost everywhere, which is incompatible with measure $1/2$.

In dimension one, choose the half-line for which the density is nonincreasing into its interior. Convexity of V guarantees one such side at the median; the density at the median is I, proving (4). For a circle with uniform measure, the two inward endpoint rays give the constant density I up to their meeting point.

3. The separated pieces

Fix $k \geq 2$, and set

$$b = \frac{1}{2k}, \qquad g = \frac{b}{k-1} = \frac{1}{2k(k-1)}. \tag{6}$$

Let

$$G(t) = \mu\{0 < f < t\}.$$

By (4), G is continuous and I-Lipschitz, with $G(0) = 0$ and $G(t) \to 1/2$ as $t \to \infty$. Choose finite numbers

$$0 = s_1 < t_1 < s_2 < t_2 < \cdots < s_{k-1} < t_{k-1}$$

such that

$$G(s_j) = (j-1)(g+b), \qquad G(t_j) = jg + (j-1)b, \qquad 1 \le j \le k-1. \tag{7}$$

All target values in (7) are below $1/2$, so such thresholds exist. Flat portions of G cause no problem, and every positive level set of f has measure zero by (4).

Define k Borel sets by

$$A_1 = \{f = 0\},$$

$$A_{j+1} = \{t_j \le f \le s_{j+1}\} \quad (1 \le j \le k-2), \qquad A_k = \{f \ge t_{k-1}\}. \tag{8}$$

For $k = 2$, the middle family is empty. The omitted gaps are

$$D_j = \{s_j < f < t_j\}, \qquad 1 \le j \le k-1,$$

and each has measure g. Therefore

$$\mu(A_1) = \frac{1}{2}, \qquad \mu(A_i) = b \quad (2 \le i \le k), \qquad \mu(A) = 1 - b, \quad A = \bigcup_{i=1}^{k} A_i. \tag{9}$$

In particular, the measure tuple belongs to the set Δ_k in the question: its sum is $1 - b$, and every individual measure is at least b.

The density bound (4) and the gap masses give

$$t_j - s_j \ge \frac{g}{I}.$$

Since f is 1-Lipschitz, any two consecutive sets in (8) are at distance at least the intervening gap width; nonconsecutive sets are farther apart. Thus

$$\min_{i \ne j} d(A_i, A_j) \ge \frac{g}{I} > 0. \tag{10}$$

4. Testing the concentration assumption

Set

$$r = \frac{g}{4I}. \tag{11}$$

By (10), this is within the radius range allowed by the hypothesis:

$$0 < r \leq \frac{1}{2} \min_{i \neq j} d(A_i, A_j).$$

If a point of a gap D_j lies in A_r, the 1-Lipschitz property of f forces its f-value to lie within r of one of that gap's two endpoints. Each such endpoint layer has measure at most Ir, by (4). Consequently

$$\begin{aligned} \mu(A_r \setminus A) &\leq 2(k-1)Ir \\ &= \frac{(k-1)g}{2} = \frac{b}{2}. \end{aligned}$$

Combining this with (9),

$$1 - \mu(A_r) \geq \frac{b}{2}. \tag{12}$$

The assumed multi-set concentration estimate, applied to the sets (8), gives

$$1 - \mu(A_r) \leq b \exp[-\min(ar^2, \sqrt{a}\, r)].$$

Together with (12), this implies

$$\min(ar^2, \sqrt{a}\, r) \leq \log 2.$$

Writing $x = \sqrt{a}\, r$, and using $\log 2 < 1$, the last inequality forces $x < 1$ and then $x^2 \leq \log 2$. Thus

$$a \leq \frac{\log 2}{r^2} = \frac{16 \log 2}{g^2} I^2 = 64 k^2 (k-1)^2 \log 2\, I^2. \tag{13}$$

5. From isoperimetry to the variational levels

The weighted Cheeger inequality on a complete manifold gives

$$\lambda^{(1)} \geq \frac{h_\mu^2}{4} = I^2. \tag{14}$$

This is a functional inequality, so it does not require an eigenfunction or a discrete spectrum. One may verify it directly: for a real $H^1(\mu)$ function u, choose a median m. Apply the coarea form of the Cheeger inequality to $(u-m)_+^2$ and $(u-m)_-^2$, whose supports have measure at most $1/2$, and then use Cauchy–Schwarz. This gives

$$\int |\nabla u|^2 d\mu \geq \frac{h_\mu^2}{4} \int |u-m|^2 d\mu \geq \frac{h_\mu^2}{4} \int |u - \mu(u)|^2 d\mu.$$

Every two-dimensional subspace in the variational definition contains a nonzero mean-zero function, proving (14). Monotonicity of the min–max levels gives $\lambda^{(k)} \geq \lambda^{(1)}$. Equations (13)–(14) prove (1).

If one permits a connected zero-dimensional manifold, it is a single point and the infimum over $(k+1)$-dimensional subspaces is empty, hence $+\infty$; the assertion is then immediate under the usual variational convention.

References and verification scope

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